

The DRanking Protocol: Verifiable Distributed Ranking and Sybil-Bounded Admission for Structured Peer-to-Peer Networks

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Abstract

DRanking is a reputation and admission protocol for open structured peer-to-peer networks. Its purpose is to convert locally observed peer behavior into a globally agreed, verifiable service weight that can gate admission to protocol roles, culminating in committee selection for Byzantine fault tolerant coordination.

The protocol is designed around one negative result and one restated assumption. The negative result, due to Douceur, is that no internal mechanism can bound the number of adversarial identities in an open network; identity creation is free. The restated assumption is therefore not “2/3 of nodes are honest” but “2/3 of accumulated, trust-weighted service is honest.” Every component of DRanking exists to make that weight expensive to forge: unforgeable signed service receipts replace unsigned counters; a seed-anchored trust propagation with conservation, attenuation, and revocation laws bounds the influence any colluding cluster can extract; verifiable random sortition replaces oracle-based sampling; a minimal proof-of-authority ledger records epoch commitments while being trusted only for ordering, never for truth; and the ledger’s state transition is a circuit-friendly pure function folded by an incrementally verifiable computation scheme, so that a browser node verifies the entire history at constant cost. Rank is soulbound and slashable; rewards are a separate transferable asset. We state the trust algebra as explicit laws and invariants, prove a bound on Sybil influence, and give an economic transition path from authority-anchored to stake-anchored operation.

I. Introduction and Motivation

An open structured peer-to-peer network admits any participant who can generate a key pair. This is a feature for reachability and a hazard for every protocol layer that must allocate a scarce resource: storage quota, relay bandwidth, committee membership, or a vote. The network needs a notion of standing — which participants have served it, and how much — that is robust against participants who manufacture identities, testimony, or both.

The 2023 draft of this protocol proposed local behavioral scoring, global aggregation by random sampling, and a median-guessing incentive game. Three years of implementation and analysis exposed a common root flaw in all three components: each assumed that honesty could be counted per identity, while identities cost nothing. Statistical validation of sample placement cannot detect an adversary who places free identities uniformly; a median game rewards whichever coalition controls half the sample; unsigned local counters aggregate into unverifiable hearsay. The present rewrite keeps the problem decomposition of the draft — local evidence, global aggregation, sampling, incentives, cold start, punishment — and replaces the adversarial assumption underneath every component.

A. Design Position

Douceur’s Sybil result [1] shows that without an external authority or a resource cost, an open system cannot bound the fraction of adversarial identities. DRanking therefore does not attempt to count honest nodes. It prices honest weight:

Assumption 1 (Honest weight). Let $t \in [0, 1]^V$ be the global trust vector defined in Section IV. DRanking assumes that identities controlled by the adversary hold strictly less than 1/3 of total trust mass:

$$\sum_{v \in \text{Adv}} t_v < \frac{1}{3} \sum_{v \in V} t_v.$$

No assumption is made about the number of adversarial identities, which is unbounded.

Every mechanism in this paper is an instrument for making Assumption 1 purchasable at a known price: the protocol composes a one-time admission cost, a continuous service cost with decay, an objective punishment channel, and (in the final phase) an external economic anchor, and the security claim is the explicit cost function of acquiring 1/3 of trust mass.

B. Contributions

- 1) Verifiable receipts (Section III): local measurement is upgraded from unsigned counters to counterparty-signed service receipts, making all aggregated evidence

third-party checkable and reducing the forgery surface to collusive self-dealing.

- 2) A trust algebra (Section IV): seed-anchored propagation with conservation, attenuation, and revocation laws; a proved upper bound on the trust mass any Sybil region can hold; soulbound, slashable rank.
- 3) Oracle-free sampling (Section V): verifiable random sortition weighted by trust, replacing the draft's decentralized random oracle and its statistical validation, with an explicit account of why the latter cannot detect Sybil density.
- 4) A minimally trusted ledger (Section VI): a small proof-of-authority BFT chain whose validators are trusted for ordering and availability only; all recorded content is self-certifying, and equivocation proofs are first-class slash objects.
- 5) A succinct chain (Section VII): the ledger state transition is a pure, circuit-friendly function; the trust fixpoint is unrolled into one power-iteration step per epoch, so the whole chain folds under incrementally verifiable computation and a browser verifies it at constant cost.
- 6) A two-asset economy (Section VIII): rank is soulbound; reward is transferable; we show why a payable rank destroys the Sybil cost anchor, and give a security-budget criterion for the authority-to-stake transition.

II. System Model and Adversary

Definition 1 (Network substrate). The substrate is a structured overlay in the Chord family [2], [3]: identities are self-certifying decentralized identifiers (DIDs) derived from public keys, placed on the circular carrier $R = \mathbb{Z}/2^{160}\mathbb{Z}$; links are authenticated pairwise transports; every message is signed by its origin. Establishing a link is cheap but not free (signaling plus session setup); creating an identity is free.

Definition 2 (Epoch). Protocol time is divided into epochs $k = 0, 1, 2, \dots$ delimited by the ledger of Section VI. All aggregation, trust updates, sortition, and reward issuance are per-epoch.

Assumption 2 (Adversary). The adversary controls an unbounded set of identities; may generate identifiers adaptively (grinding); may deviate arbitrarily from the protocol (Byzantine); may collude across all its identities, including exchanging valid signatures among them; and may participate honestly for arbitrarily long before deviating (patience). The adversary cannot forge signatures of identities it does not control and cannot break the verifiable random function of Section V.

The patience clause matters: any reputation earned by behavior can be farmed by an adversary willing to behave. DRanking's response is not to prevent farming but to price it (continuous cost, Section III), decay it (Law 3), and make a single detected equivocation forfeit it entirely (Section VI).

III. Verifiable Service Receipts

The draft aggregated locally maintained counters (connect/disconnect, send/receive successes and failures). Such counters are first-person testimony: sound as local link-selection evidence, but unverifiable the moment they are reported to a third party. A global ranking built on them aggregates hearsay. DRanking makes the unit of evidence a signed receipt.

Definition 3 (Service receipt). A receipt is a tuple

$$\rho = \langle \text{srv}, a, b, k, n, \sigma_b \rangle$$

in which srv names a service kind (relay, storage challenge, liveness probe), a is the serving DID, b the served DID, k the epoch, n a nonce unique per (a, b, k) pair scope, and σ_b the signature of b over the preceding fields. A receipt is valid if σ_b verifies and (b, k, n) has not been seen before.

Proposition 1 (Forgery surface). Under Assumption 2, an adversary cannot fabricate a valid receipt naming an honest b as signer. The only manufacturable receipts are those signed within the adversary's own identity set (self-dealing).

Self-dealing is not prevented at this layer; it is neutralized at the trust layer, where a receipt's contribution is weighted by the signer's trust (Section IV). A cluster of zero-trust identities signing receipts to one another injects approximately zero weight.

Definition 4 (Receipt graph). The evidence of epoch k is a weighted directed multigraph aggregated into the matrix $W_k \in \mathbb{R}_{\geq 0}^{V \times V}$, where $W_k[b, a]$ is the service value receipted by signer b to server a during epoch k , under a per-signer, per-epoch issuance cap $W_k[b, \cdot] \mathbf{1} \leq w_{\max}$.

The issuance cap makes receipt-signing itself a budgeted act: a signer cannot certify more service than the protocol deems one identity able to consume in one epoch. This is the first of several conservation-shaped constraints; the trust layer adds the second.

Receipts price honest weight continuously: to hold weight, an identity must keep producing service that real counterparties keep countersigning, which costs bandwidth and uptime for as long as the weight is to be held. The protocol deliberately certifies performed service rather than proving capacity with synthetic load; manufactured traffic between colluding identities would burn network capacity and still be discounted to zero by signer-trust weighting.

IV. The Trust Algebra

Global standing is a vector $t_k \in [0, 1]^V$ computed from receipt graphs and a seed vector. We state the algebra as laws, because each law closes a specific attack.

Definition 5 (Seed vector). A seed vector $s \in [0, 1]^V$, $\sum_v s_v = 1$, assigns initial trust to a small authority set S (Section VI). Seeds are the protocol's explicit trust root:

DRanking does not derive trust from nothing, it declares where trust originates and then constrains how it flows.

Law 1 (Conservation). Trust is redistributed, never created. The propagation matrix C_k is the row-substochastic normalization of W_k : $C_k[b, a] = W_k[b, a] / \max(w_{\max}, W_k[b, \cdot] \mathbf{1})$. A signer's total outflow of endorsement can never exceed its own unit row, and endorsement given is budget spent.

Law 2 (Attenuation). Trust decays along every hop by a damping factor $\alpha \in (0, 1)$: a chain of endorsements of length ℓ carries at most α^ℓ of its source weight.

Law 3 (Revocation and decay). Trust is a flow, not a stock: it must be re-earned every epoch because the update law re-derives t_{k+1} from current receipts; a slashed identity (Section VI) is removed from the carrier of t and its held mass returns to the seed term; seeds may withdraw endorsement.

Definition 6 (Trust update).

$$t_{k+1} = \alpha C_k^T t_k + (1 - \alpha) s. \quad (1)$$

Equation (1) is one step of power iteration toward the personalized-PageRank/EigenTrust fixpoint [4], [5] $t^* = (1 - \alpha)(I - \alpha C^T)^{-1} s$. DRanking deliberately does not compute the fixpoint within an epoch: one step per epoch is executed, so the fixpoint is approached along chain time. This choice is semantic as well as computational — standing accrues over history, not instantaneously — and it is what makes the transition function circuit-friendly in Section VII.

Invariant 1 (Mass). If $\sum_v t_0(v) = 1$ then for all k , $\sum_v t_k(v) \leq 1$, with equality when every row of C_k is fully spent. Trust mass never inflates.

Proposition 2 (Sybil influence bound). Partition V into an honest region H (including all seeds) and an adversarial region A . Let the attack inflow at epoch k be $g_k = \sum_{b \in H, a \in A} C_k[b, a] t_k(b)$, the trust-weighted endorsement flowing from honest signers to adversarial servers, and let $g = \sup_k g_k$. Then for all k ,

$$T_k(A) = \sum_{v \in A} t_k(v) \leq \frac{\alpha g}{1 - \alpha}.$$

Proof. Since no seed lies in A , the seed term contributes nothing to $T_{k+1}(A)$. Splitting the propagation term by origin region,

$$T_{k+1}(A) \leq \alpha(g_k + T_k(A)),$$

because mass arriving in A from A itself is at most $\alpha T_k(A)$ by Law 1 (rows are substochastic, so internal circulation cannot amplify) and mass arriving from H is exactly αg_k . Iterating from $T_0(A) = 0$ gives a geometric series bounded by $\alpha g \sum_{i \geq 0} \alpha^i = \alpha g / (1 - \alpha)$. $\square$

The bound has the right shape for engineering: adversarial standing is linear in honest endorsement actually granted to the adversary (g), and independent of how

many identities the adversary creates. Defense concentrates on making g small and expensive: endorsement costs budget (Law 1), honest signers only countersign service they received, and detected misbehavior claws standing back (Law 3).

Definition 7 (Soulbound rank). The rank of v at epoch k is $t_k(v)$. Rank is bound to the DID that earned it: it cannot be transferred, delegated, or sold; it decays under Law 3; and it is forfeited entirely upon an accepted misbehavior proof. Section VIII shows that relaxing non-transferability re-opens the Sybil market the rest of the protocol closes.

A. Cold Start and Dilution

At genesis, $t_0 = s$: all standing is seed standing. As the network serves and receipts accumulate, the propagation term grows relative to the seed term. The protocol commits, in advance, to a dilution schedule: the effective seed weight $(1 - \alpha)$ in Equation (1) is a non-increasing function of cumulative receipted service, so authority influence declines on a published curve rather than by promise. The same schedule governs the consensus transition in Section VIII.

V. Sampling by Verifiable Sortition

Global aggregation requires selecting, per epoch, which nodes' local receipt sets are collected and which nodes audit the epoch commitment. The draft selected by lookup toward targets derived from a “decentralized random oracle,” then validated the sample with entropy and Kolmogorov–Smirnov tests. Both halves fail, for instructive reasons.

A. Why Statistical Validation Cannot Work

The draft's tests check that sampled positions are uniformly placed. Uniformity of placement is orthogonal to honesty of identity: an adversary who mints free identifiers uniformly over R occupies any target fraction of the identifier space while passing every placement test exactly as an honest population would. Statistical tests detect biased sampling; the threat is cheap identity. No test on positions can see the difference, so validation must instead be cryptographic (receipts, signatures, proofs) and economic (trust weighting). Separately, a global random seed is a bootstrap circularity: a decentralized randomness beacon already requires the committee that sampling is meant to select.

B. Sortition

DRanking removes the global oracle. Each node holds a verifiable random function (VRF) [6] key bound to its DID. For epoch k with finalized ledger beacon B_k (the hash of the finalized epoch- k header, unpredictable before finality under Assumption 2):

Algorithm 1 Trust-weighted sortition (per node v , epoch k)

- 1: $(y, \pi) \leftarrow \text{VRF}_{\text{sk}_v}(B_k \parallel \text{role})$
 - 2: if $y/2^{|y|} < \tau \cdot t_k(v)$ then v holds role for epoch k , with proof π
-

Selection probability is proportional to rank, so Assumption 1 translates directly into an honest-majority statement about every sampled set: sortition is the functor from trust mass to committee seats. Grinding is closed at both inputs: the VRF key is fixed before B_k exists, and B_k is fixed by finality before the adversary can steer it. Self-selection with public proofs also removes the draft’s median game entirely — there is no reported quantity for a coalition to steer, hence no game whose equilibrium must be argued.

VI. The Admission Ledger

Aggregated evidence needs a single ordered record: without one, a node can show different evidence to different observers, and no deterministic global rank exists. DRanking uses a deliberately small ledger.

A. Minimal Trust Design

Definition 8 (Authority ledger). A BFT proof-of-authority chain operated by the seed set S with instant finality (two-thirds quorum signatures on each block). Per epoch, the chain records: the epoch commitment (a Merkle root of W_k), the resulting trust vector commitment (root of t_{k+1}), validator-set changes, slash records, and reward issuance. Raw receipt data remains in the overlay’s storage layer; the chain holds commitments.

Obligation 1 (Validators order, content certifies). Every recorded object must be independently verifiable from signatures and commitments alone: receipts carry their signers’ signatures, trust vectors are recomputable from committed inputs by Equation (1), and slashes carry proofs. A colluding validator quorum can censor or re-order; it must never be able to fabricate. Any proposed ledger content that a validator quorum could forge is a design defect at this layer.

Censorship, the residual validator power, is bounded by a force-inclusion rule: a node holding a validator-acknowledged submission that does not appear on chain within a timeout publishes the acknowledgment as a misbehavior proof, slashing the withholding quorum’s members.

B. Equivocation as the Objective Signal

All positive evidence (receipts) is verifiable but collusive; the only fully objective reputation events are negative:

Definition 9 (Equivocation proof). Two messages, both validly signed by the same DID, asserting contradictory protocol statements for the same slot (two epoch commitments for the same epoch, two votes for one height,

two receipts with one nonce). The pair is self-certifying: verifying it requires no honesty assumption at all.

Equivocation proofs are first-class ledger objects: anyone may submit one; inclusion slashes the offender’s rank to zero (Law 3) and, in the staked phase, its bond. This is what defeats the patient adversary of Assumption 2: months of farmed standing are forfeited by one detected contradiction, so the expected value of farming falls with the certainty of eventual detection.

VII. Succinct State by Folding

The substrate’s defining deployment target is the browser: a wasm node with neither the storage to replay history nor a reason to trust a validator signature it cannot audit. DRanking therefore requires the ledger to be succinctly verifiable: a constant-size proof attests the validity of the entire chain state.

A. The Step Function

Definition 10 (Epoch transition). The ledger state is $\Sigma_k = \langle \text{root}(t_k), \text{root}(V_k), \text{root}(\text{slash}_k), \text{root}(\text{bal}_k) \rangle$. The transition $F(\Sigma_k, \text{root}(W_k)) = \Sigma_{k+1}$ is a pure function: one application of Equation (1), slash application, validator-set update, and reward issuance, over Merkle-committed inputs.

Because Equation (1) is a single sparse matrix–vector product with scalar damping — additions and multiplications over committed data — F is circuit-friendly by construction. This is the payoff of unrolling the fixpoint: the draft’s alternative (an eigenvector solve per epoch) is hostile to arithmetization; one power-iteration step is trivial under it. The design rule generalizes:

Obligation 2 (Circuit discipline). Every extension of the ledger state machine must keep F pure, deterministic, and expressible in cheap arithmetic over committed inputs. The ledger is an admission ledger, not a general contract platform; each added semantic multiplies proving cost and freezes harder under Obligation 3.

B. Folding

F is folded by an incrementally verifiable computation scheme of the Nova family [7]: each epoch folds its step instance into a running accumulator; a compressed proof π_k attests “ Σ_k is reachable from Σ_0 by k valid applications of F .” A fresh browser node verifies π_k plus the finality quorum signature on Σ_k — constant work, after which it trusts mathematics for validity and the quorum only for canonicity.

Two structural facts make the combination clean. First, IVC folds a linear chain of steps; BFT instant finality guarantees the chain never reorganizes, so no re-folding branch logic exists — probabilistic-finality chains cannot say this. Second, proving is asynchronous: the chain advances by replay verification among validators (cheap, since F is pure), while the prover compresses history

behind the head; succinctness is a property of verification, and lags nothing but itself.

Obligation 3 (Upgrade versioning). Under IVC the circuit is the constitution: any change to F changes the verifier key. The protocol carries an explicit version register in Σ ; upgrades bind old and new verifier keys at a committed epoch boundary. This mechanism ships at genesis, not as a retrofit.

What folding does not provide: canonicity (consensus chooses the chain), data availability (the overlay stores receipt data), or privacy of the receipt graph. The granularity of what $\text{root}(W_k)$ exposes — full graph, per-node aggregates, or blinded commitments — is an open design decision with surveillance consequences, and zero-knowledge openings of individual standing are its natural extension.

VIII. Incentives and Economics

A. Two Assets, Not One

The draft made rankings “payable.” This is the single most consequential error in the 2023 design:

Proposition 3 (Payable rank defeats cost anchoring). If rank is transferable at market price p , the adversary’s cost of acquiring trust mass m is $\min(\text{farm}(m), p \cdot m)$, and honest exiting nodes are rational sellers; the market clears at the price of the cheapest accumulated history, decoupling standing from behavior. Non-transferability (Definition 7) removes the second argument of the min, restoring $\text{farm}(m)$ — service, time, and slash risk — as the only acquisition path.

DRanking therefore carries two assets: rank, soulbound, earned by receipted service, governing admission and sortition weight; and reward, a transferable token issued per epoch to nodes in proportion to receipted, trust-weighted service. Reward is the distribution mechanism: by the time the network transitions to stake, token holders are approximately its service providers, because service was the only faucet.

B. Value and Sinks

A reward token distributed against service is worthless without a sink. The protocol’s fee loop makes the token the payment instrument for the network’s scarce services: storage persistence beyond baseline replication, relay quality-of-service, and namespace registration. Fees are burned or recycled into the epoch reward pool. The security consequence is direct: the stake anchor of the next subsection is only as strong as the demand routed through these sinks, so the fee loop is a security parameter, not a business afterthought.

C. The Authority-to-Stake Transition

The ledger launches as proof-of-authority (Section VI) because quorum arithmetic requires a closed set and

genesis has no other Sybil anchor. The transition to proof-of-stake replaces the fiat validator set with a sortition over bonded token weight. Two rules govern it:

Obligation 4 (Security-budget crossover). The switch is conditioned, not scheduled: stake-based validation activates only when slashable bonded value exceeds a published multiple of the estimated attack payoff. Before the crossover, PoA finality signatures remain a hybrid overlay on PoS block production, weighted down along the same dilution schedule as the seed vector (Section IV).

Obligation 5 (One abstraction boundary). Consensus is parameterized over an abstract validator-set function $\text{vset} : \Sigma_k \rightarrow 2^V$; PoA supplies a constant, PoS supplies a bonded-sortition function. The switch changes one pure function, not the engine.

Proof-of-work alternatives were evaluated and rejected: hash-rental markets price a majority attack on a young chain at hours of rented compute; and “proof of traffic” fails twice — synthetic load between colluding identities is indistinguishable from service (so it buys no Sybil resistance) while consuming the very bandwidth the network exists to provide. Receipted real service with trust weighting is the version of proof-of-traffic that keeps the evidence and discards the waste.

IX. Security Analysis

Table I summarizes the threat matrix. Each row names the attack, the mechanism that bounds it, and the residual exposure.

Table I
Threat matrix

Attack	Bounding mechanism	Residual
Sybil flood	Prop. 2: mass bounded by honest endorsement, not identity count	$\alpha g / (1 - \alpha)$
Self-dealing receipts	Signer-trust weighting (Prop. 1); issuance caps	zero-trust circulation
Patient farming	Continuous cost, decay (Law 3), total slash on equivocation	undetected deviation
Whitewashing	Fresh DIDs start at $t = 0$; standing is slow to earn, instant to lose	re-entry at zero
Sampling grind/eclipse	VRF sortition on finalized beacon; keys precede beacon	—
Validator censorship	Force-inclusion with acknowledgment proofs; slashing	latency
Seed compromise	Threshold multi-seed endorsement; revocation; dilution schedule	pre-dilution window
Rank market	Soulbound (Prop. 3) rank	out-of-band key sales

Two residuals deserve emphasis. Out-of-band key sales (selling the DID’s private key, rank attached) cannot be

prevented cryptographically; the mitigation is economic — a sold identity carries its slash exposure, and a buyer inherits a history it did not control. The pre-dilution window is the honest name for the PoA phase: until the dilution schedule and the security-budget crossover retire them, the seeds are a threshold trust root, and the paper claims exactly that, no more.

X. Related Work

DRanking’s local layer descends from behavior scoring in file-sharing networks — eDonkey’s ranking queues [8] and BitTorrent’s choking [9] — which the 2023 draft adopted directly. EigenTrust [5] and personalized PageRank [4] supply the propagation fixpoint; DRanking’s contribution is unrolling it along chain epochs and binding it to signed evidence. The social-graph Sybil defenses SybilGuard, SybilLimit, and SybilRank [10]–[12] established the attack-edge bound pattern that Proposition 2 instantiates over a service graph. Byzantine-resilient peer sampling (Brahms [13]) addresses the sampling-under-adversary problem that the draft’s statistical tests could not; DRanking sidesteps it with Algorand-style VRF sortition [6], [14] weighted by trust instead of stake. S/Kademlia [15] prices identifier creation with puzzles, a complementary one-time cost. The ledger design follows PBFT-lineage instant finality [16]; the succinct chain follows Mina [17], realized with Nova-family folding [7]. The impossibility frame is Douceur’s [1]; the Byzantine quorum frame is Lamport, Shostak, and Pease’s [18]; the economic transition draws on Nakamoto’s observation that external cost, not internal testimony, anchors open membership [19].

XI. Conclusion

The 2023 draft decomposed the reputation problem correctly and secured each component against the wrong adversary — one who lies, rather than one who is legion. This rewrite keeps the decomposition and re-anchors every component in cost: receipts make testimony expensive to forge, the trust algebra makes influence expensive to accumulate and instant to lose, sortition makes selection impossible to steer, the ledger makes history impossible to equivocate, folding makes verification cheap enough for a browser, and the two-asset economy keeps standing off the market. The protocol’s honest claim is a priced assumption: the system is safe while one third of trust mass costs more than attacking it pays. Every mechanism above is a term in that price.

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